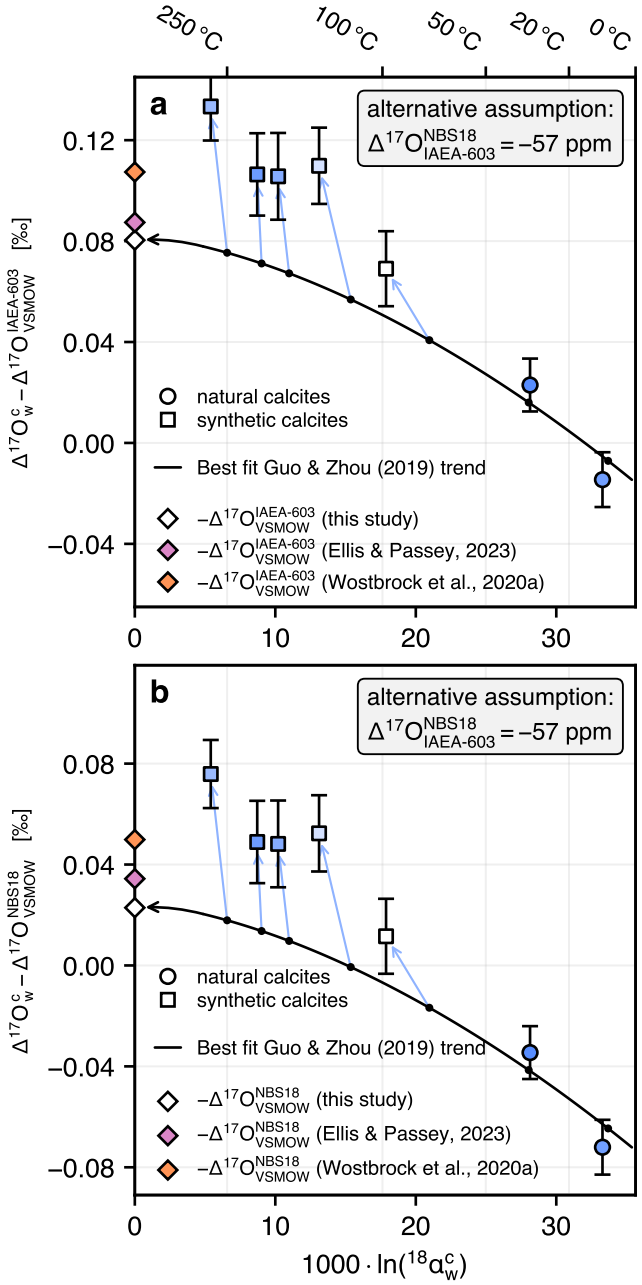
250 °C
100 °C
50 °C
20 °C
0 °C
a
alternative assumption:
$\Delta^{17}O^{NBS18}_{IAEA-603} = -57$ ppm
$\Delta^{17}O^{c}_{w} - \Delta^{17}O^{IAEA-603}_{VSMOW}$ [‰]
0.12
0.08
0.04
0.00
−0.04
natural calcites
synthetic calcites
Best fit Guo & Zhou (2019) trend
$-\Delta^{17}O^{IAEA-603}_{VSMOW}$ (this study)
$-\Delta^{17}O^{IAEA-603}_{VSMOW}$ (Ellis & Passey, 2023)
$-\Delta^{17}O^{IAEA-603}_{VSMOW}$ (Wostbrock et al., 2020a)
0
10
20
30
b
alternative assumption:
$\Delta^{17}O^{NBS18}_{IAEA-603} = -57$ ppm
$\Delta^{17}O^{c}_{w} - \Delta^{17}O^{NBS18}_{VSMOW}$ [‰]
0.08
0.04
0.00
−0.04
−0.08
natural calcites
synthetic calcites
Best fit Guo & Zhou (2019) trend
$-\Delta^{17}O^{NBS18}_{VSMOW}$ (this study)
$-\Delta^{17}O^{NBS18}_{VSMOW}$ (Ellis & Passey, 2023)
$-\Delta^{17}O^{NBS18}_{VSMOW}$ (Wostbrock et al., 2020a)
0
10
20
30
$1000 \cdot \ln(^{18}\alpha^{c}_{w})$